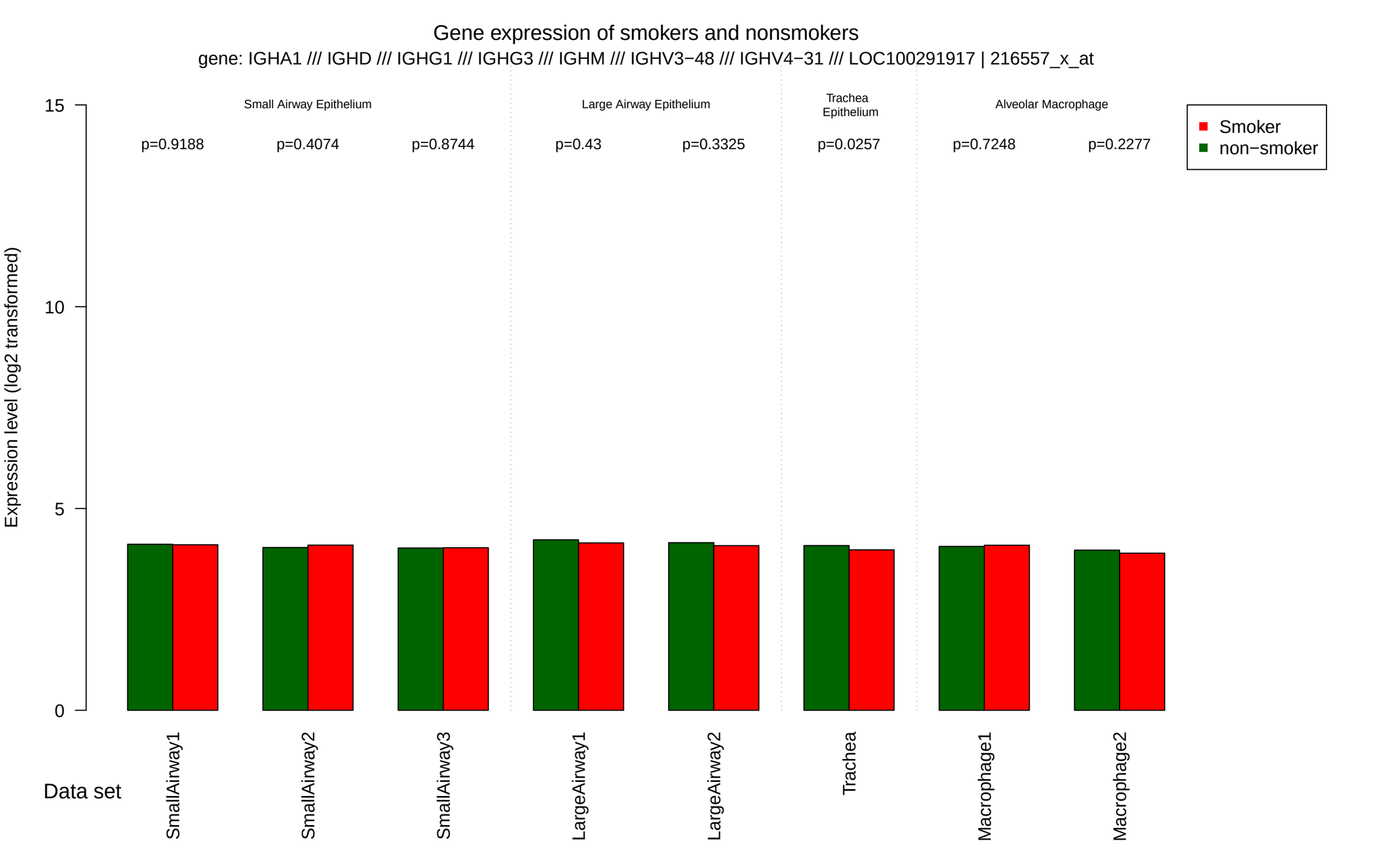
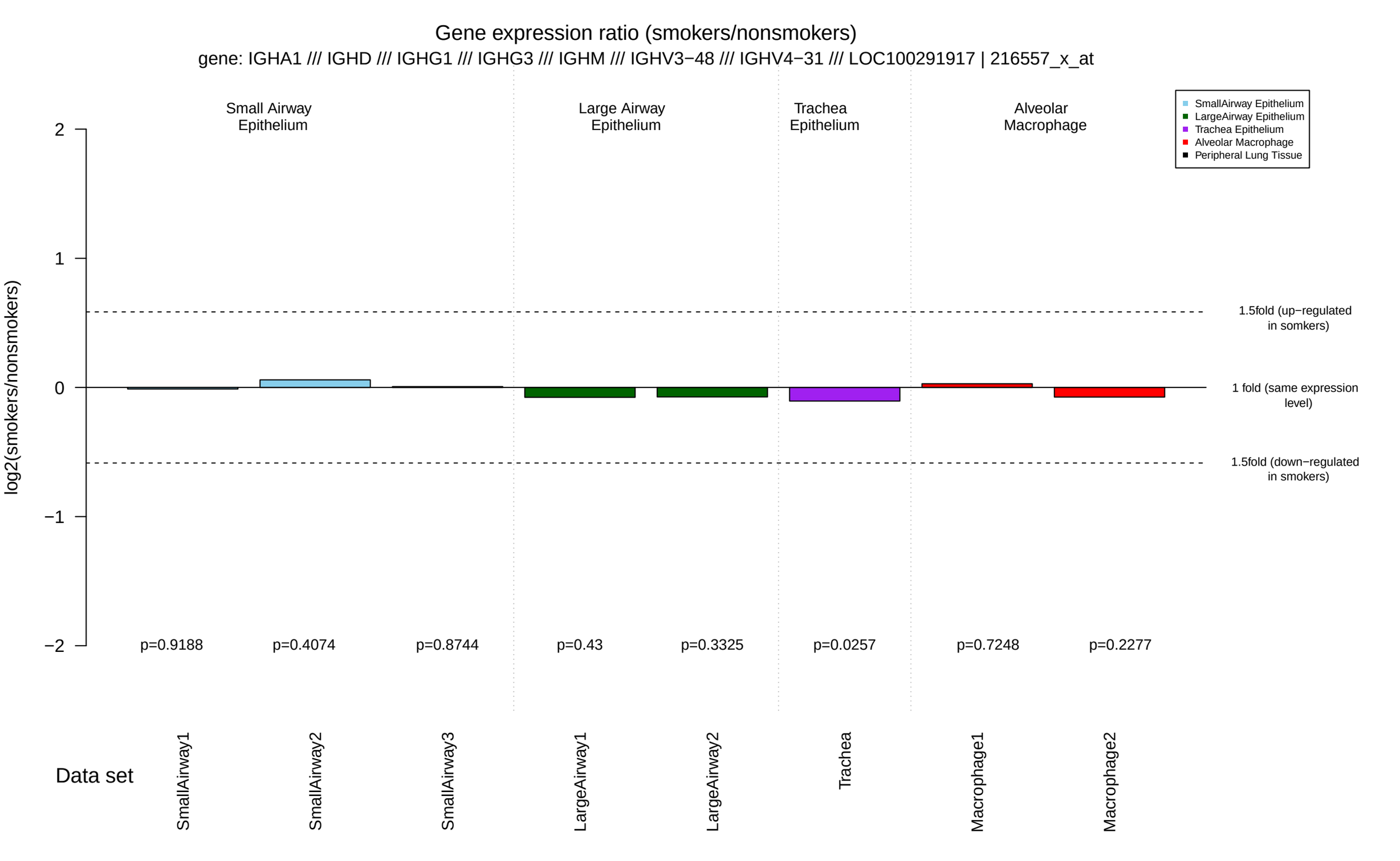
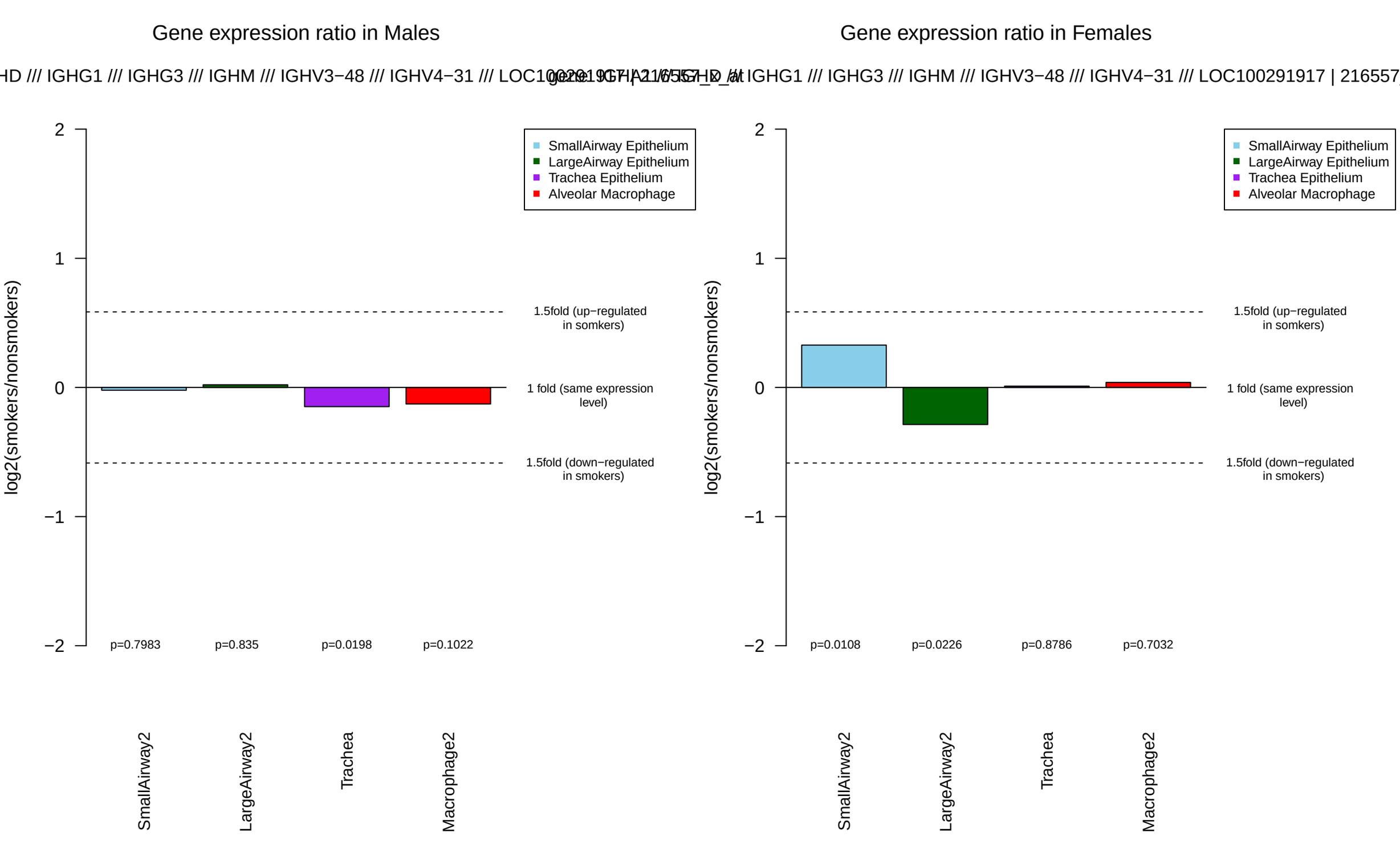
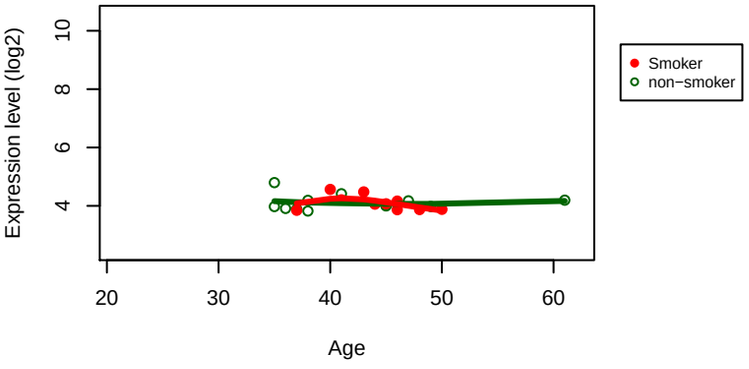

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC100291917 | 216557_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9188
p=0.4074
p=0.8744
p=0.43
p=0.3325
p=0.0257
p=0.7248
p=0.2277
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

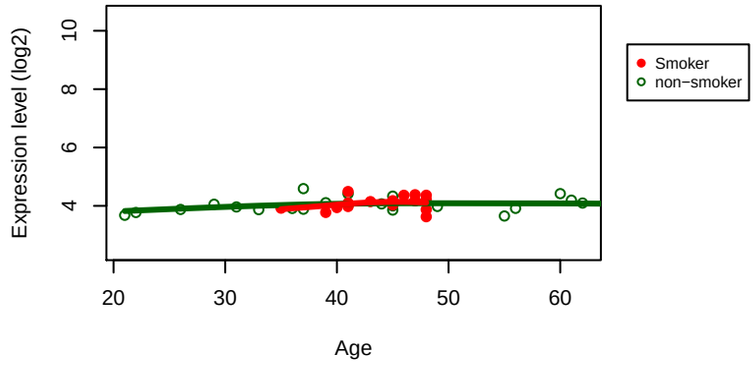

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC100291917 | 216557_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9188
p=0.4074
p=0.8744
p=0.43
p=0.3325
p=0.0257
p=0.7248
p=0.2277
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio in Males
HD /// IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC100291917 | 216557_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7983
p=0.835
p=0.0198
p=0.1022
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC100291917 | 216557
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0108
p=0.0226
p=0.8786
p=0.7032
SmallAirway2
LargeAirway2
Trachea
Macrophage2

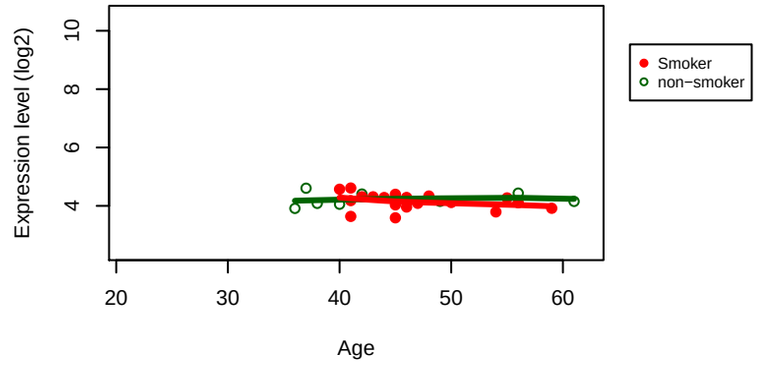
set: SmallAirway1 (p=0.9188)



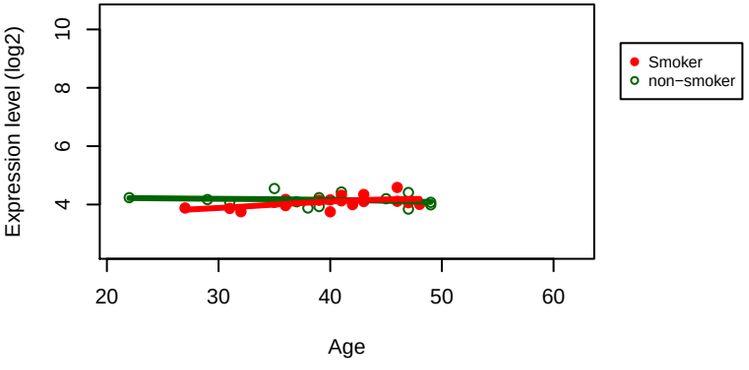
set: SmallAirway2 (p=0.4074)



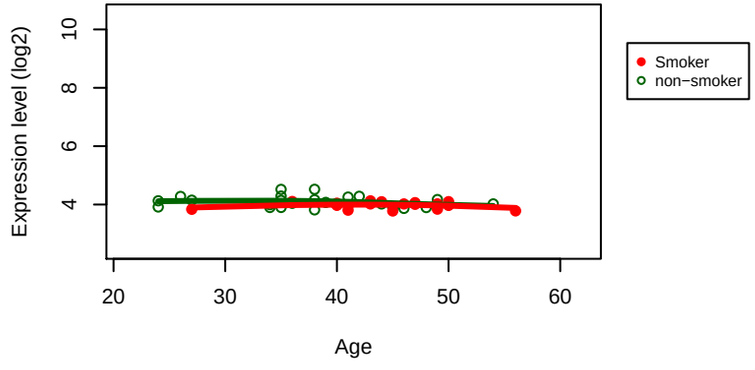
set: LargeAirway1 (p=0.43)



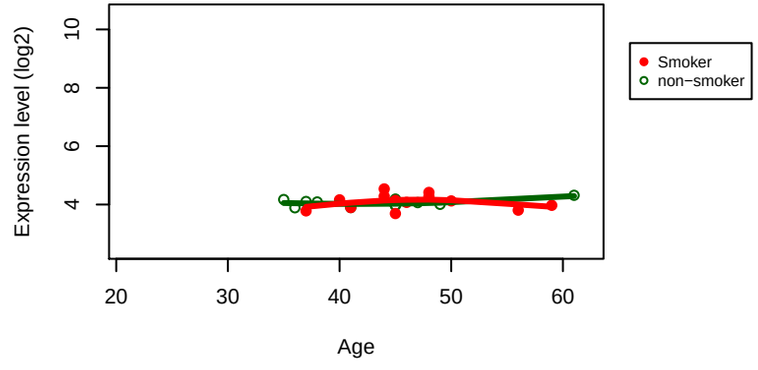
set: LargeAirway2 (p=0.3325)



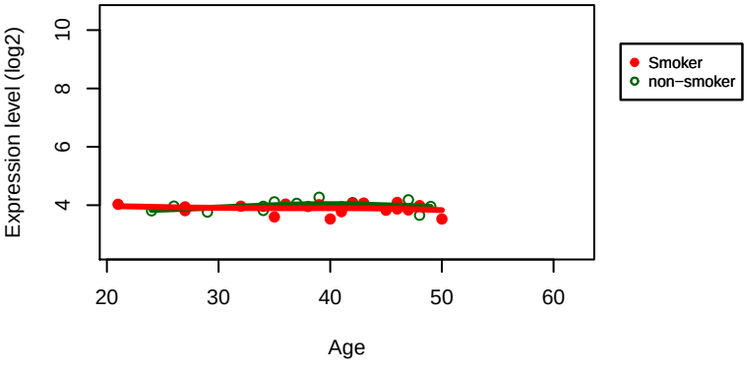
set: Trachea (p=0.0257)

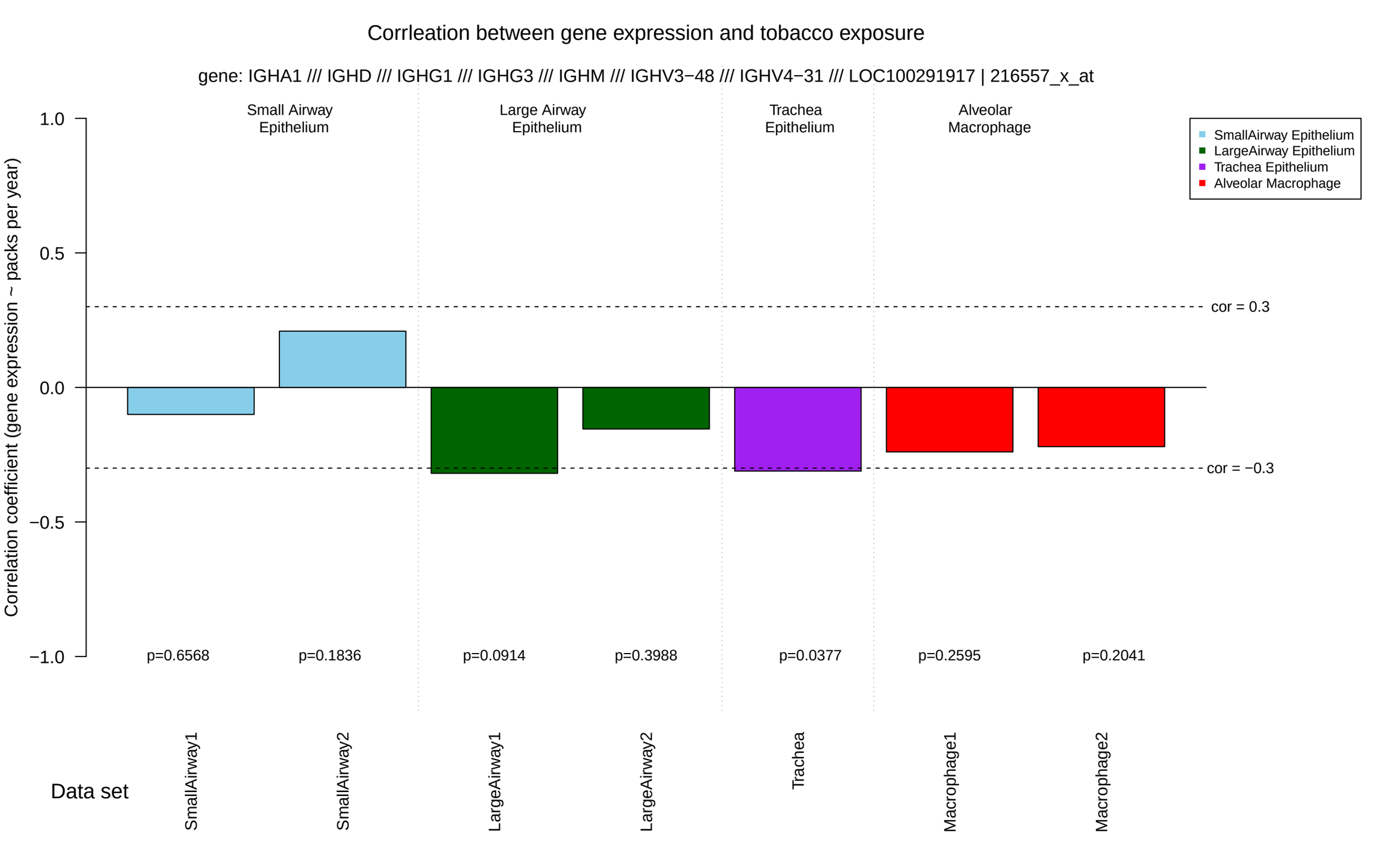


set: Macrophage1 (p=0.7248)



set: Macrophage2 (p=0.2277)




Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV3−48 /// IGHV4−31 /// LOC100291917 | 216557_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6568
p=0.1836
p=0.0914
p=0.3988
p=0.0377
p=0.2595
p=0.2041
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2